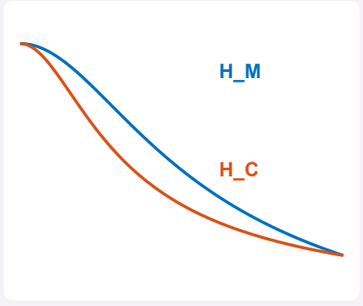
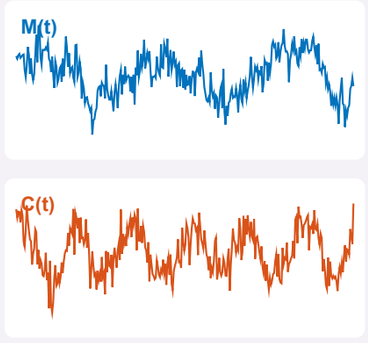
1. Fix one family



Input spectra and true pathways
remain fixed within the family

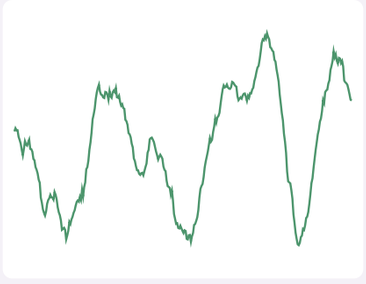
2. Generate inputs



Shared and independent random
fluctuations create the inputs

3. Create clean CBFV

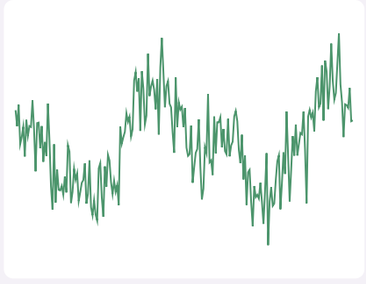
V_clean = h_M * M + h_C * C



The generator stores h_M and h_C
as the pathway truth

4. Create an observation

Finite duration + noise + timing



The observation changes
while the true system does not

5. Estimate and compare

Observed M, C, and V

Estimated H_M and H_C

Compare the estimates with
the stored true H_M and H_C

The estimator never receives truth

One family fixes the generating system; observation and estimator choices test how accurately it can be recovered.